Pratt truss

Plane truss — analysis and CIRSOC 301 rod checks

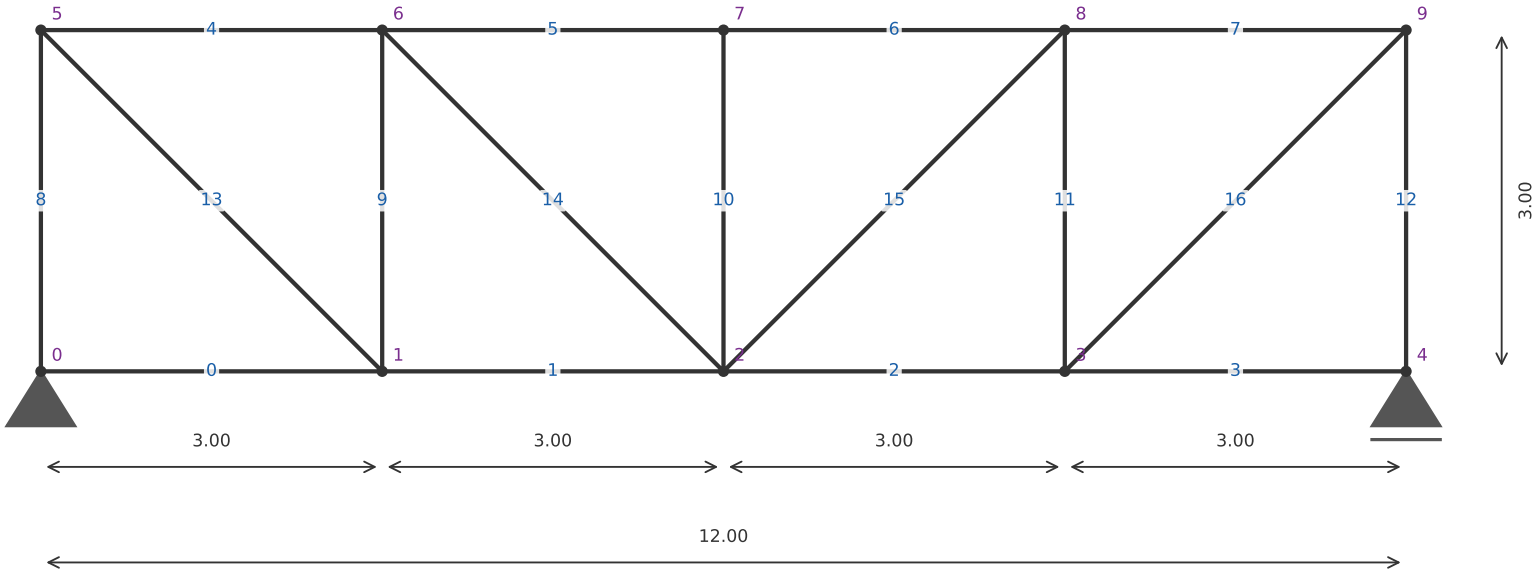
Span	12.00 m
Height	3.00 m
Nodes / rods	10 / 17
Supports	pin at 0, rollerX at 4
Loads	3 nodal, total 120.0 kN
Family "Chord"	CHS 114.3x4.5 (F24 (CIRSOC) / A36)
Family "Diagonal"	CHS 60.3x3.6 (F24 (CIRSOC) / A36)
Family "Vertical"	CHS 88.9x4.0 (F24 (CIRSOC) / A36)
Weight	Weighs 513 kg, carries 120.0 kN (24 × its own weight), passes (0.63).
Deflection	10.25 mm = L/1171, limit L/300: OK
Stability	Statically determinate (m + r − 2j = 17 + 3 − 2×10 = 0): statics alone gives every force.
Most used rod	rod 16 at 0.63 (normal stress (H.3.4))
Rods over capacity	0 of 17 checked
Buckling	$\lambda_1 = 3.55$
Shear and moment	none: every rod is pin-ended and loaded only at its nodes, so the axial force is the whole result

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9. Explain rod 16 and 13

Results are an aid to a qualified professional who reviews them. Rods modelled as pin-ended members; geometric first-order analysis; joints and connections are not checked here.

Node numbers in purple, rod numbers in blue. Dimensions in m.



2 m · true to scale

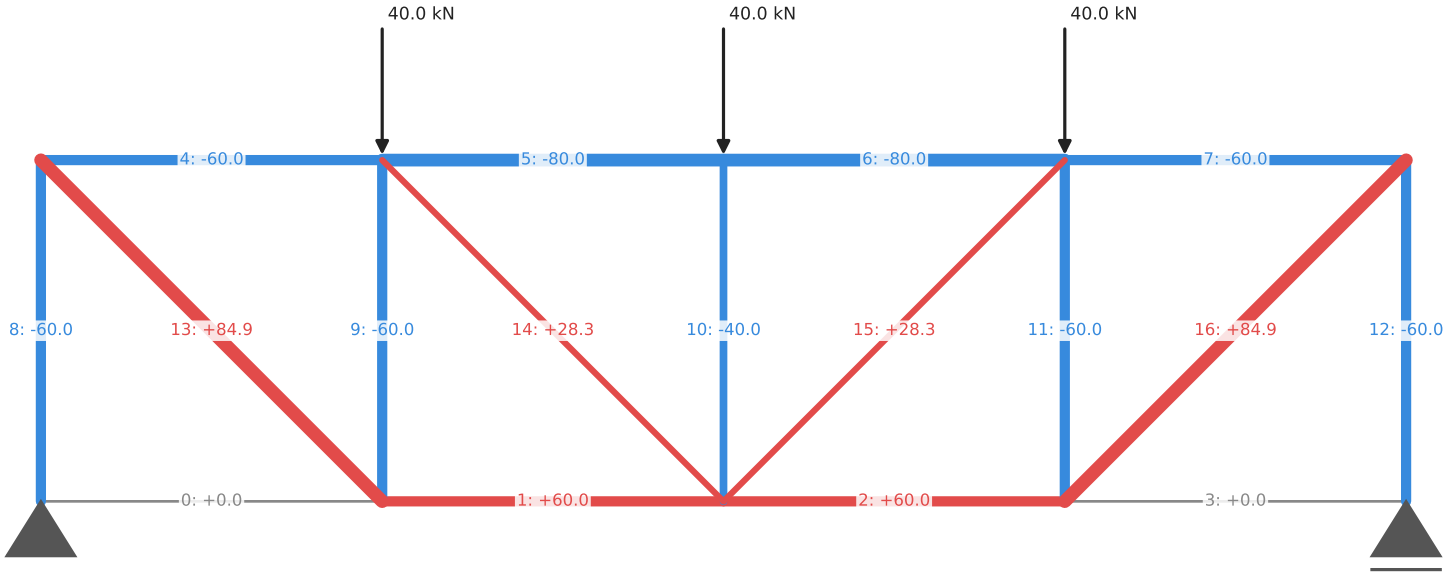
0 2

Truss — Geometry

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MODEL	SIZE	UNITS	CODE	DATE
Pratt truss	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Axial force N (kN): red pulls (+), blue pushes (−), grey carries nothing; line width follows |N| (largest 84.9).



2 m · true to scale

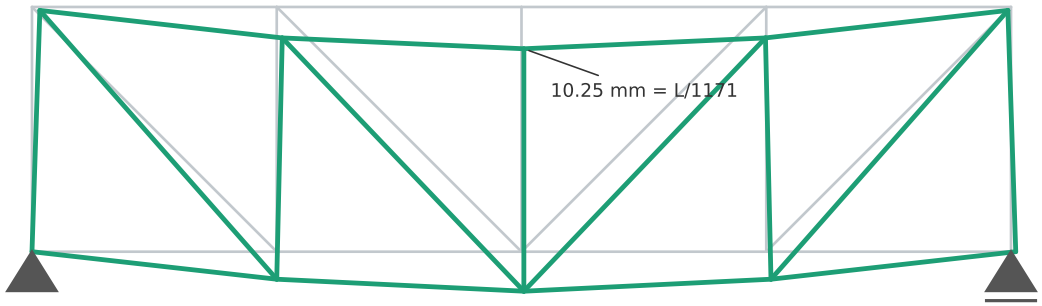
0 2

Truss — Axial forces

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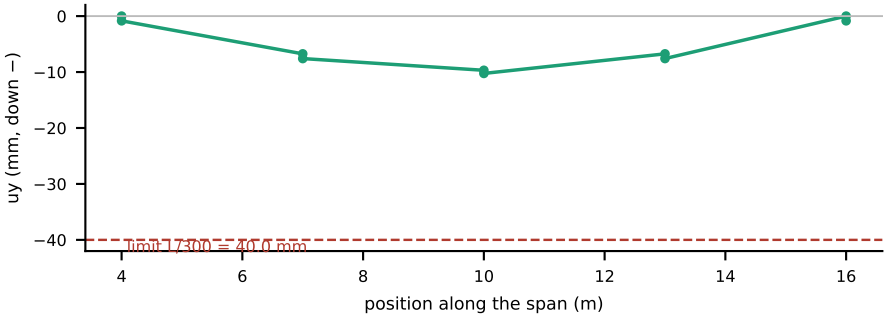
MODEL	SIZE	UNITS	CODE	DATE
Pratt truss	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Deflected shape (green) over the truss (grey). Geometry to scale; displacements magnified × 50. Largest vertical deflection 10.25 mm against the limit L/300 = 40.0 mm: OK.



Elastic deformation

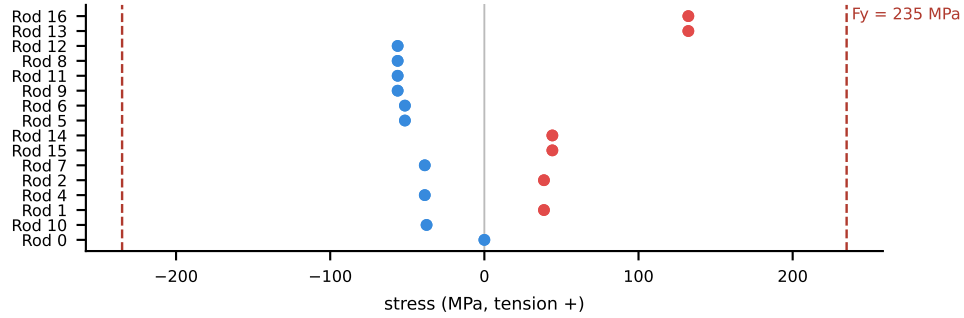
vertical movement of every joint



Largest 10.25 mm at joint 7 = L/1171. Within the limit.

Fibre stress

N/A ± M·c/I at each rod's extreme fibres



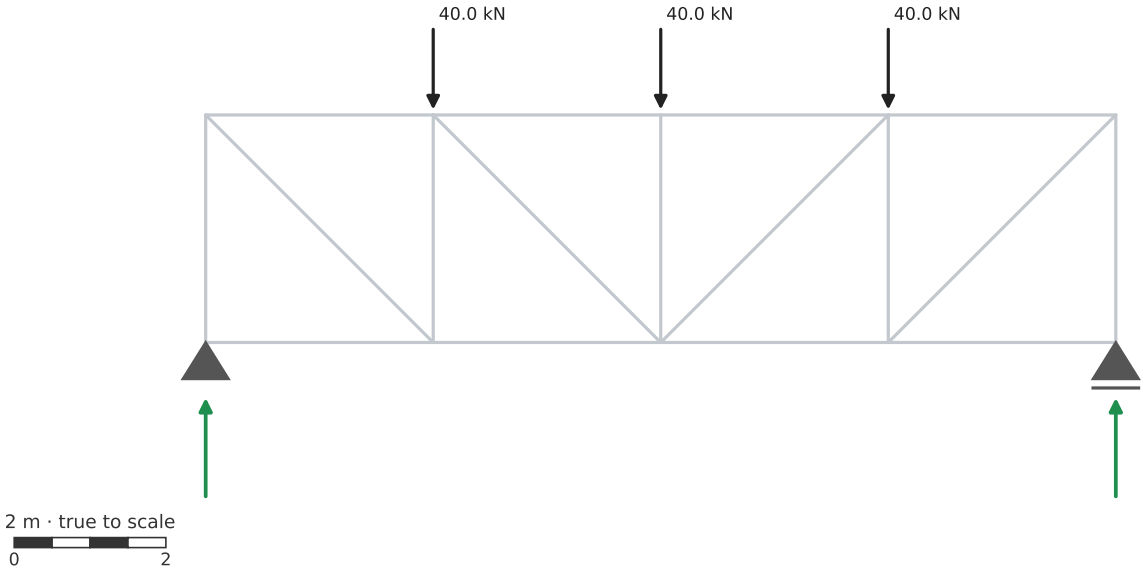
Every rod is pinned, so its two fibres carry the same N/A and coincide: no bending opens the pair.

Truss — Deflection and stress

MODEL	SIZE	UNITS	CODE	DATE
Pratt truss	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Support reactions (green arrows), as the supports push on the truss; Ry positive upwards. The sums balance the loads.

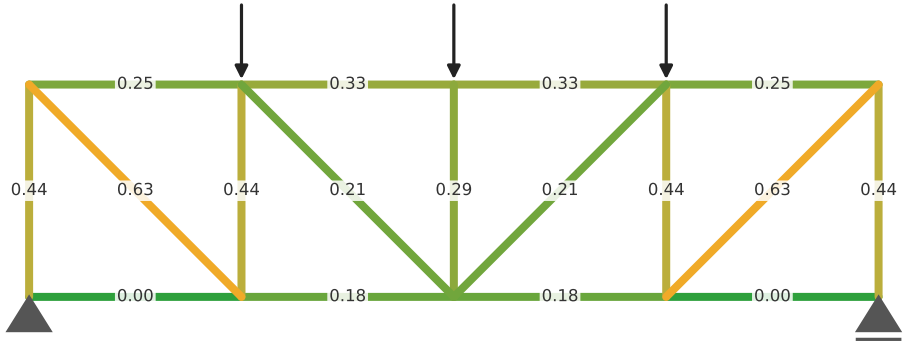
Node	Type	Rx (kN)	Ry (kN)
0	pin	0.00	60.00
4	rollerX	0.00	60.00
Σ		0.00	120.00



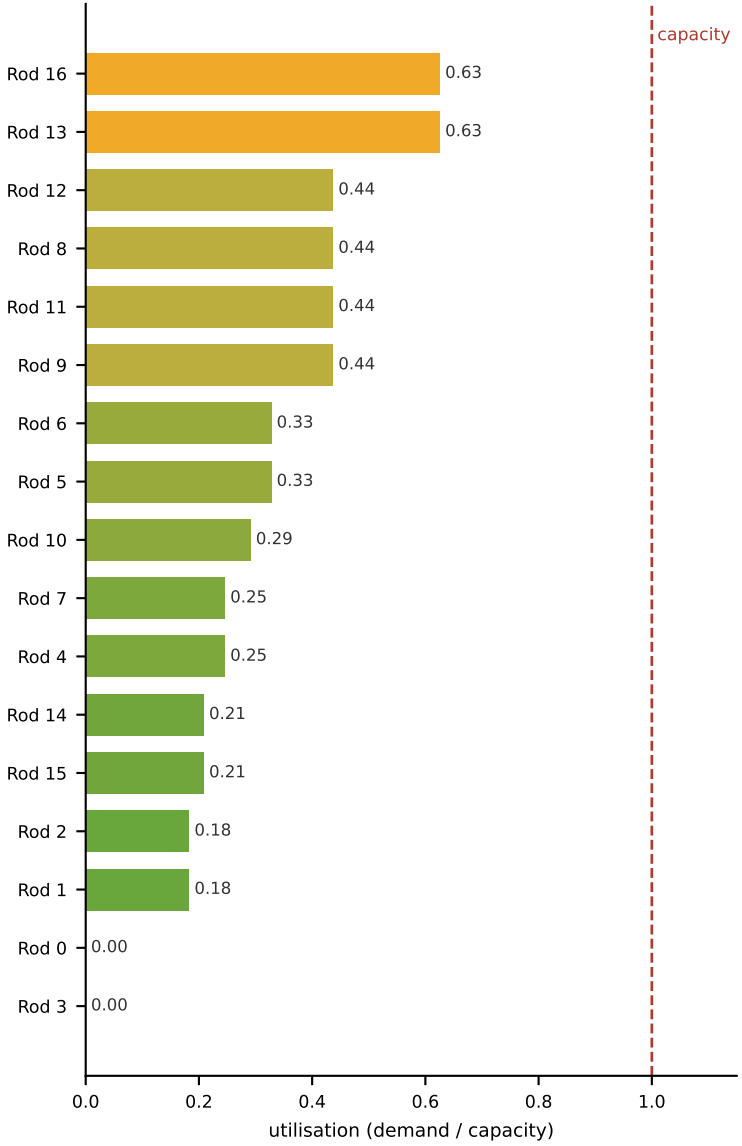
Truss — Reactions

MODEL	SIZE	UNITS	CODE	DATE
Pratt truss	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Utilisation: demand over CIRSOC 301 capacity. Most used rod 16 at 0.63 (normal stress (H.3.4)). Every checked rod is within capacity.



2 m · true to scale
0 2





grey = no steel section, not checked

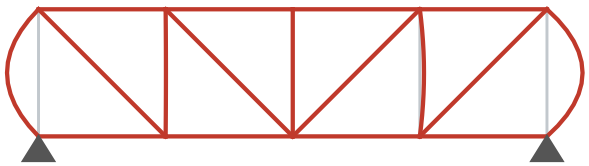
Truss — Utilisation

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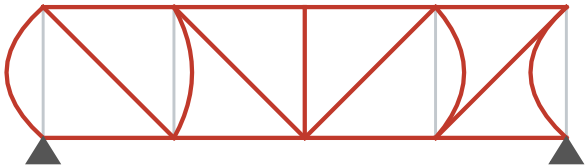
MODEL	SIZE	UNITS	CODE	DATE
Pratt truss	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

λ is the factor on the load as it stands. $\lambda_1 = 3.55$, so the load can grow 3.5× before it buckles. 3 modes share that λ: that many equal members buckle at the same load, so stiffening one of them buys nothing.

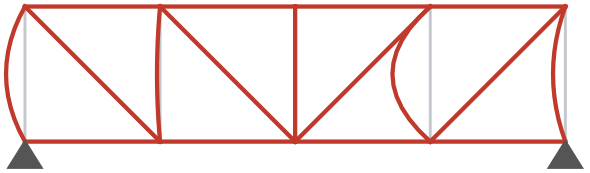
Mode 1: λ = 3.55
rod 12 bows

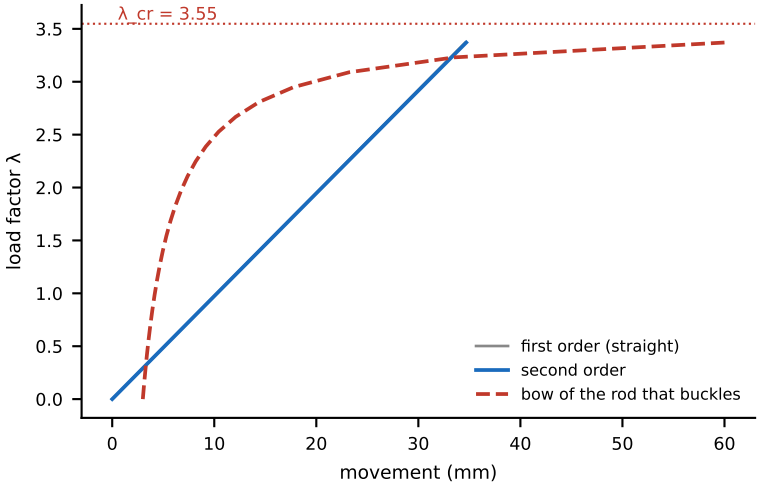


Mode 2: λ = 3.55
rod 8 bows



Mode 3: λ = 3.55
rod 11 bows





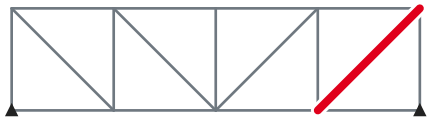
The second-order line bends away from the first-order one and runs off as λ nears λ_cr: that gap IS the amplification every second-order design rule is about. Started from an initial bow of L/1000 = 3.00 mm in the first mode's shape.

Elastic buckling of the IDEAL truss, so every λ is an UPPER bound: imperfections, residual stresses and yielding are not modelled, and each rod is cut in two so it can bow between its joints. Buckling OUT of the plane is the rod check's job (CIRSOC 301 E3, minor radius) and is in the Utilisation and Rods sheets, not here. A mode has a shape, not a size: the shapes are drawn exaggerated.

Truss — Buckling

MODEL	SIZE	UNITS	CODE	DATE
Pratt truss	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use	
0	0–1	Chord	CHS 114.3x4.5	3.00	+0.00	inelastic buckling (E3-2)	0.00	OK
1	1–2	Chord	CHS 114.3x4.5	3.00	+60.00	normal stress (H.3.4)	0.18	OK
2	2–3	Chord	CHS 114.3x4.5	3.00	+60.00	normal stress (H.3.4)	0.18	OK
3	3–4	Chord	CHS 114.3x4.5	3.00	+0.00	normal stress (H.3.4)	0.00	OK
4	5–6	Chord	CHS 114.3x4.5	3.00	-60.00	inelastic buckling (E3-2)	0.25	OK
5	6–7	Chord	CHS 114.3x4.5	3.00	-80.00	inelastic buckling (E3-2)	0.33	OK
6	7–8	Chord	CHS 114.3x4.5	3.00	-80.00	inelastic buckling (E3-2)	0.33	OK
7	8–9	Chord	CHS 114.3x4.5	3.00	-60.00	inelastic buckling (E3-2)	0.25	OK
8	0–5	Vertical	CHS 88.9x4.0	3.00	-60.00	inelastic buckling (E3-2)	0.44	OK
9	1–6	Vertical	CHS 88.9x4.0	3.00	-60.00	inelastic buckling (E3-2)	0.44	OK
10	2–7	Vertical	CHS 88.9x4.0	3.00	-40.00	inelastic buckling (E3-2)	0.29	OK
11	3–8	Vertical	CHS 88.9x4.0	3.00	-60.00	inelastic buckling (E3-2)	0.44	OK
12	4–9	Vertical	CHS 88.9x4.0	3.00	-60.00	inelastic buckling (E3-2)	0.44	OK
13	5–1	Diagonal	CHS 60.3x3.6	4.24	+84.85	normal stress (H.3.4)	0.63	OK
14	6–2	Diagonal	CHS 60.3x3.6	4.24	+28.28	normal stress (H.3.4)	0.21	OK
15	8–2	Diagonal	CHS 60.3x3.6	4.24	+28.28	normal stress (H.3.4)	0.21	OK
16	9–3	Diagonal	CHS 60.3x3.6	4.24	+84.85	normal stress (H.3.4)	0.63	OK



Rod 16

utilisation 0.63

normal stress (H.3.4)

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Rod 16
======
Steel, CIRSOC 301-2018 -- A = 641 mm², Fy = 235 MPa, L = 4.243 m

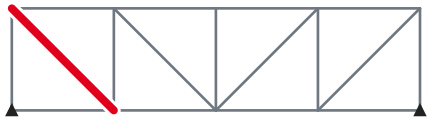
 1. Axial stress
      σ = N / A
      = 84,853 N / 641 mm²
      = 132.32 MPa

 2. Design stress  [H.3.4]
      φ · Fy
      = 0.9 × 235
      = 211.50 MPa

 3. Tension ratio  [H.3.4]
      σ / (φ Fy)
      = 132.32 / 211.50
      = 0.626

 4. Governs
      the largest ratio
      = normal stress (H.3.4)
      = 0.626

utilisation 0.626 -- OK
```



Rod 13

utilisation 0.63

normal stress (H.3.4)

```
Rod 13
======
Steel, CIRSOC 301-2018 -- A = 641 mm², Fy = 235 MPa, L = 4.243 m

 1. Axial stress
      σ = N / A
      = 84,853 N / 641 mm²
      = 132.32 MPa

 2. Design stress  [H.3.4]
      φ · Fy
      = 0.9 × 235
      = 211.50 MPa

 3. Tension ratio  [H.3.4]
      σ / (φ Fy)
      = 132.32 / 211.50
      = 0.626

 4. Governs
      the largest ratio
      = normal stress (H.3.4)
      = 0.626

utilisation 0.626 -- OK
```